EnviroPredict

Air Quality Index (AQI) Prediction System

Comprehensive Technical & Visual Report

1. Executive Summary

EnviroPredict is a hyper-local, machine learning-powered Air Quality Index (AQI) prediction system designed specifically for the state of Odisha. Combining real-time environmental data pipelines with an advanced **XGBoost ML Engine**, the application delivers highly accurate predictions across 9 districts and over 400 locations.

MODEL ACCURACY	DATA POINTS	SUPPORTED LOCATIONS
98.83%	**25,000+**	**400+**

2. AQI Levels & Color System

The system categorizes air quality into six distinct levels, following standard environmental guidelines. Each level is associated with a specific color hex code used consistently across the UI for gauge meters, charts, and alerts.

AQI RANGE	CATEGORY	COLOR HEX	VISUAL INDICATOR
0 - 50	**Good**	`#22c55e`	Good
51 - 100	**Satisfactory**	`#eab308`	Satisfactory
101 - 200	**Moderate**	`#f97316`	Moderate
201 - 300	**Poor**	`#ef4444`	Poor

AQI RANGE	CATEGORY	COLOR HEX	VISUAL INDICATOR
301 - 400	**Very Poor**	`#a855f7`	Very Poor
401 - 500	**Severe**	`#dc2626`	Severe

3. Pollutant Parameters Analyzed

The prediction model evaluates six critical pollutants along with temperature and humidity. The user interface visualizes these pollutants with dedicated thematic colors and dynamic progress bars based on their safe maximum limits.

PM2.5

Unit: **$\mu g/m^3$** | Max Safe Scale: 250

PM10

Unit: **$\mu g/m^3$** | Max Safe Scale: 430

Ozone (O_3)

Unit: **$\mu g/m^3$ / ppb** | Max Safe Scale: 200

Nitrogen Dioxide (NO_2)

Unit: **$\mu g/m^3$ / ppb** | Max Safe Scale: 200

Carbon Monoxide (CO)

Unit: **mg/m^3** | Max Safe Scale: 10

Sulfur Dioxide (SO_2)

Unit: **$\mu g/m^3$ / ppb** | Max Safe Scale: 350

4. Model Performance & Metrics

The system's performance is comprehensively evaluated using both regression and classification metrics to ensure utmost reliability of the AQI predictions.

REGRESSION METRICS

R^2 Score: 0.9961

RMSE: 3.41

MAE: 1.40

CLASSIFICATION METRICS

Accuracy: 98.30% (0.9830)

Precision: 0.9828

Recall: 0.9830

F1-Score: 0.9829

CROSS-VALIDATION

CV R^2 Scores: [0.9754, 0.9833, 0.9917, 0.9891, 0.9360]

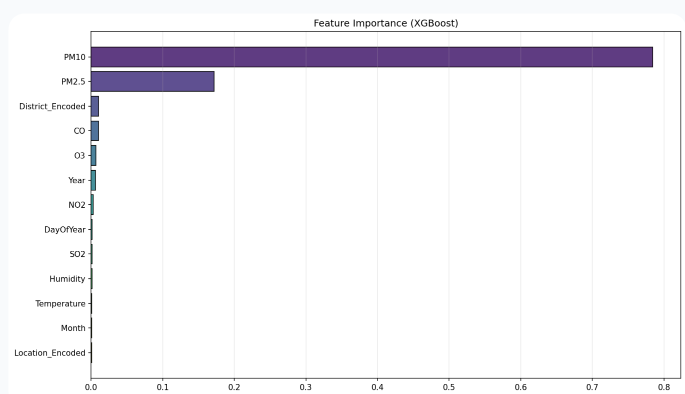
Mean CV R^2: 0.9751 ± 0.0203

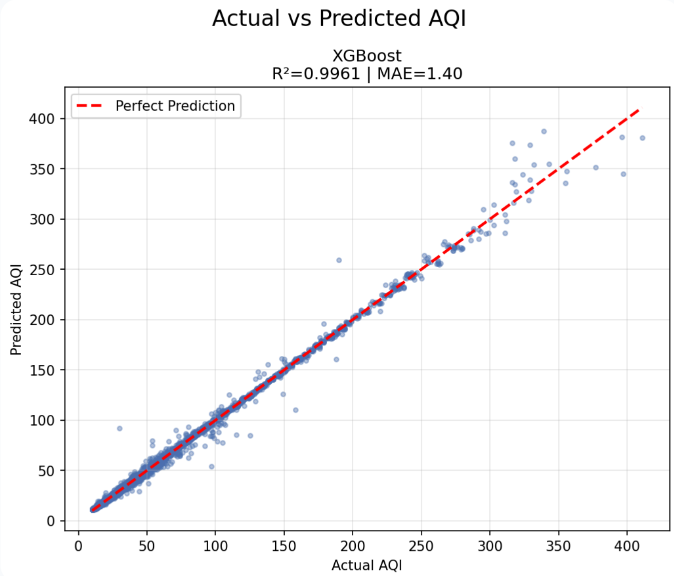
5. Data Visualizations & Analysis

A comprehensive set of plots generated during Exploratory Data Analysis (EDA) and Model Training phases.

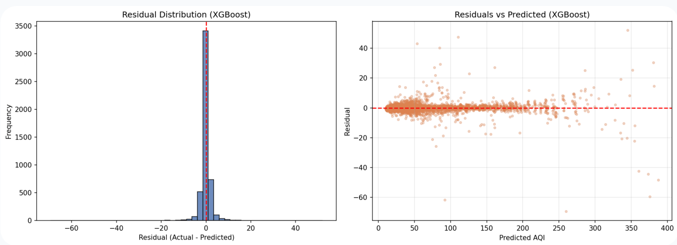
ACTUAL VS PREDICTED AQI

FEATURE IMPORTANCE

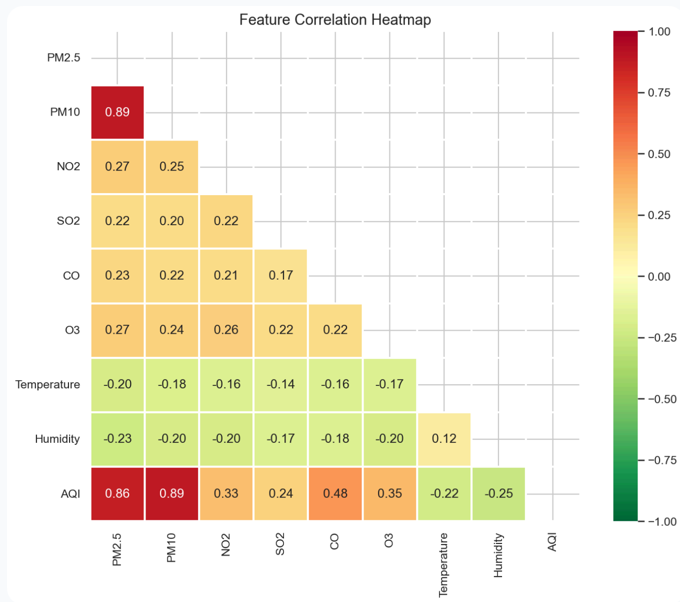




RESIDUAL ANALYSIS

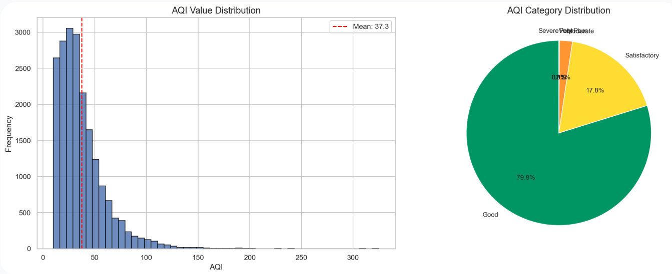


CORRELATION HEATMAP

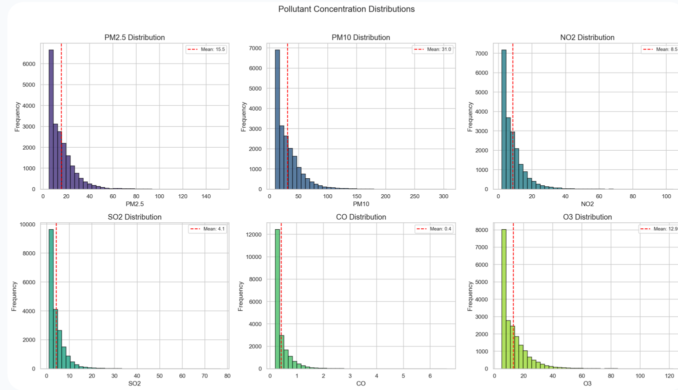


Distribution & Trends

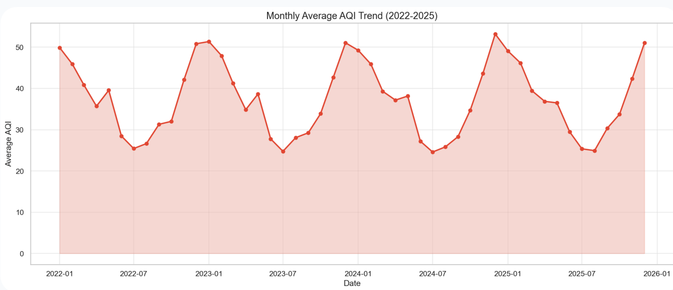
AQI DISTRIBUTION



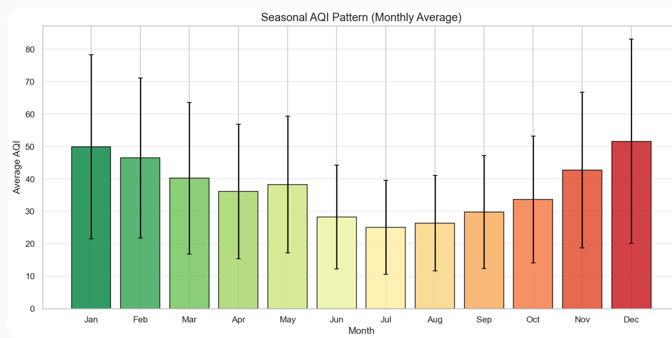
POLLUTANT DISTRIBUTIONS



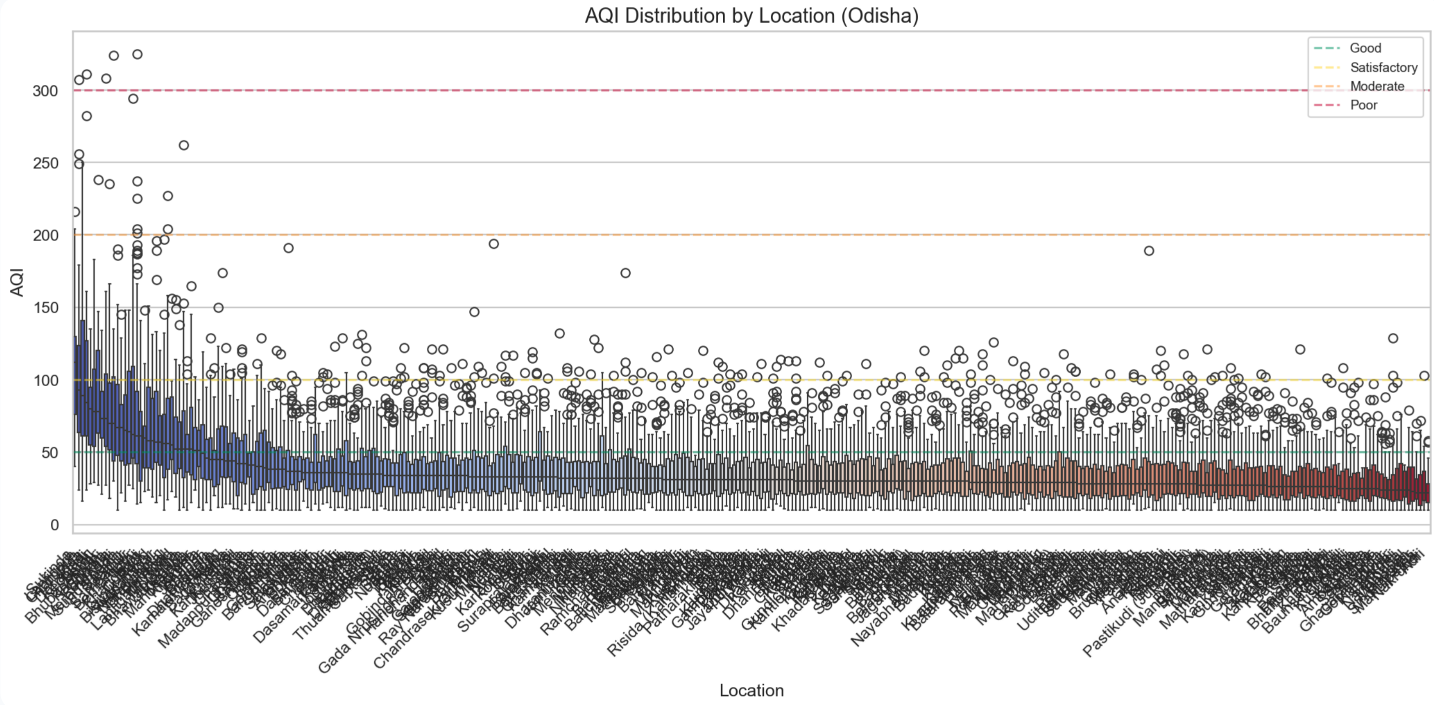
MONTHLY AQI TREND



SEASONAL AQI PATTERN



LOCATION WISE AQI


AQI Distribution by Location (Odisha)
Good
Satisfactory
Moderate
Poor
AQI
0
50
100
150
200
250
300
Location

6. User Interface & Experience (UI/UX)

The EnviroPredict dashboard features a hyper-responsive, cinematic design aimed at delivering complex environmental data through a highly accessible and visually striking interface.

BENTO GRID ARCHITECTURE

A modern, compartmentalized 3-column dashboard layout that elegantly organizes scan controls, dynamic pollutant breakdowns, and the main ML predicted gauge meter into distinct semantic blocks.

HYPER-RESPONSIVE SCALING

Built with a mobile-first approach using custom media queries, the application flawlessly adapts across all devices, scaling from sweeping desktop vistas to perfectly stacked mobile column layouts.

DYNAMIC DARK/LIGHT THEMING

Users can seamlessly toggle between Light and Dark modes. The application intelligently adjusts all background cards, text contrast, glow effects, and interactive elements to minimize eye strain.

POLISHED MICRO-INTERACTIONS

Premium hover lift effects, staggering element fade-ups, and native CSS transitions make the data exploration process feel alive, premium, and highly engaging.

7. Technology Stack

- XGBoost (ML Engine)
- FastAPI (REST Backend)
- React (Frontend UI)
- Vite (Build Tool)
- scikit-learn (Data Pipeline)
- Open-Meteo (Live AQ Data)
- Tailwind / Vanilla CSS
- Lucide Icons

8. Development Team

Shubhranshu Behera
ML & BACKEND LEAD

Rupak Ranjan Parida
DATA ANALYSIS

Ranjan Kumar Nayak
RESEARCH

Pramod Kumar Mohananta
FRONTEND

EnviroPredict AQI Prediction System | Built with care in Odisha | Date: 2026-05-02